

# Evanns Morales-Cuadrado

U.S. Citizen

**Robotics PhD Candidate**  
**Fulbright Specialist**  
**Georgia Institute of Technology**

## CONTACT

**Email:** [evannsmc@gmail.com](mailto:evannsmc@gmail.com)

**GitHub:** <https://github.com/evannsmc>

**Research Website:** <https://www.evannsmc.com/>

**LinkedIn:** <https://www.linkedin.com/in/evanns-morales/>

## TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, C#/.NET, Matlab/Simulink
- **Robotics & Control Frameworks:** ROS 2, NATS, PX4, ArduPilot, acados, CasADi
- **Machine Learning & Simulation:** JAX, PyTorch, MuJoCo, Reinforcement Learning
- **Development & DevOps Tools:** Docker, Continuous Integration, Git/GitHub
- **Hardware & Experimentation:** UAV assembly, repair, and deployment
- **Motion-Capture:** OptiTrack and Vicon
- **Languages:** Spanish (Native), English (Native), Portuguese (B1)

## EDUCATION

### Georgia Institute of Technology

2022-Present

Ph.D. in Robotics

*Doctoral Minor in Optimization Methods*

- GPA: 4.0/4.0
- Advisor: Dr. Samuel Coogan
- Thesis Proposal Approved (ABD)
- Goizueta Fellow

### University of Texas at Arlington

2018-2022

Honors Bachelor of Science in Electrical Engineering

*Minor in Mathematics*

*Minor in Physics*

*Certificate in Unmanned Vehicle Systems*

- GPA: 3.9/4.0 Summa Cum Laude
- Full National Merit Scholarship Awarded
- Electrical Engineering Honors Scholar

## WORK EXPERIENCE

### Year-Round Graduate Research Intern | AI for Autonomy

May 2026 - Present

Sandia National Laboratories

Albuquerque, NM

- Applied reinforcement learning to autonomy problems in support of national security
- Designed and deployed robotics and automation infrastructure that enables multi-robot experiments
- Built C#/.NET networking frameworks and execution scripts that coordinate communication and task execution across multi-robot systems, now used team-wide

### Graduate Research Assistant | FACTS Lab

August 2022 - Present

Georgia Institute of Technology

Atlanta, GA

- Advised by Dr. Samuel Coogan
- Developed a lightweight Newton-Raphson tracking controller that beat PX4's native tracker in aggressive quadrotor flight and matched or beat NMPC on blimp and quadrotor hardware at far lower computation time and CPU energy
- Built the lab's real-time flight-test stack on PX4, ROS 2, Raspberry Pi companion computers, and Vicon/OptiTrack motion capture, enabling repeatable hardware validation of autonomy algorithms
- Developed safe planning and control methods that combine reachability analysis, formal methods, optimization, and learning to generate fast, provably safe trajectories under unknown disturbances

- Teaching Assistant** | Vertically Integrated Project in Robotics January 2024 - May 2026  
Georgia Institute of Technology Atlanta, GA
- Mentored about 20 undergraduates per semester on hands-on robotics research, with multiple mentees going on to robotics PhD programs
  - Designed and delivered undergraduate lectures on ROS, Git/GitHub, Linux, and control theory, preparing students for research on hardware
- Research Assistant** | Autonomous Systems Lab November 2021 - May 2022  
University of Texas at Arlington Arlington, TX
- Advised by Dr. Frank Lewis
  - Contributed to reinforcement-learning quadrotor control research under Dr. Frank Lewis, then ranked 12<sup>th</sup> in the USA and 23<sup>rd</sup> worldwide
  - Completed UTA Honors College thesis on lightweight Newton-Raphson tracking control for computationally constrained aerial systems
- Teaching Assistant** | Graduate-Level Intelligent Systems Course January 2022 - May 2022  
University of Texas at Arlington Arlington, TX
- Selected as an undergraduate to TA a graduate-level course
  - Led exam review sessions and graded homework, exams, and projects
- Research Assistant** | Dynamical Networks and Control Lab October 2019 - May 2022  
University of Texas at Arlington Arlington, TX
- Advised by Dr. Yan Wan
  - Conducted original research on learning-based minimum-time and minimum-energy multi-vehicle path planning using ROS and OpenCV

## SELECTED PUBLICATIONS

- E. Morales-Cuadrado**, S. Coogan, Z. Kakish, and A. Parikh. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*. (title redacted until the double-blind review process is complete)
- L. Pinto\*, **E. Morales-Cuadrado**\*, S. Coogan, M. A. Wehrmeister, L. Caselatto, and C. E. Magrin. Submitted to *2027 IEEE International Conference on Robotics and Automation (ICRA)*. \*Equal contribution. (title redacted until the double-blind review process is complete)
- M. Cao, A. Harapanahalli, **E. Morales-Cuadrado**, and S. Coogan, “Trajectory Tracking for Systems with Unknown Time-Varying Disturbances.” *IEEE Open Journal of Control Systems (OJ-CSYS)*, 2026.
- E. Morales-Cuadrado**, “Computationally Efficient and Safe Control for Aerial Robotic Systems under Threat and Disturbance.” Ph.D. Thesis Proposal, Georgia Institute of Technology, 2026. [pdf available](#). [slides available](#).
- E. Morales-Cuadrado**, L. Chung, S. Kousik, and S. Coogan, “RTD-RAX: Fast, Safe Trajectory Planning for Systems under Unknown Disturbances.” Joint Submission to 2026 Modeling, Estimation and Control Conference (MECC) and ASME Letters in Dynamic Systems and Control (ALDSC). (accepted to both) [preprint available](#).
- E. Morales-Cuadrado**, L. Baird, Y. Wardi, and S. Coogan, “Lightweight Tracking Control for Computationally Constrained Aerial Systems with the Newton-Raphson Method.” *IEEE Transactions on Control Systems Technology (TCST)*, 2026. [preprint available](#).
- L. Baird, **E. Morales-Cuadrado**, and S. Coogan, “Runtime Assurance for Uncertain Systems from Interval Signal Temporal Logic.” Submitted to IFAC Nonlinear Analysis: Hybrid Systems (NAHS), 2025. (under revision) [preprint available](#).
- E. Morales-Cuadrado**, C. Llanes, Y. Wardi, and S. Coogan, “Newton-Raphson Flow for Aggressive Quadrotor Tracking Control.” *2024 American Control Conference (ACC)*. [preprint available](#).

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|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>RESEARCH INTERESTS</b>        | <ul style="list-style-type: none"> <li>• Safe Autonomy</li> <li>• Hardware Deployment</li> <li>• Unmanned Ground and Aerial Vehicles</li> </ul>                                                                                                                                                                                                                                                                                                                                        | <ul style="list-style-type: none"> <li>• Advanced Nonlinear Control</li> <li>• Learning-Based Control</li> <li>• Trajectory Generation and Planning</li> </ul> |
|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                |
| <b>OPEN-SOURCE CONTRIBUTIONS</b> | <ul style="list-style-type: none"> <li>• <b>PX4 Autopilot:</b> Contributed documentation improvements to the official PX4 Autopilot manual.</li> <li>• <b>Vicon4PX4:</b> Developed and open-sourced a ROS 2 C++ Vicon–PX4 interface enabling external vision fusion for indoor quadrotor flight experiments; adopted and showcased by Georgia Tech’s Indoor Flight Lab as part of its official codebase. Also created a parallel package for OptiTrack systems.</li> </ul>             |                                                                                                                                                                |
| <b>SERVICE &amp; LEADERSHIP</b>  | <b>Co-Founder and President of Puerto Rican Student Association</b> August 2024 - Present<br>Georgia Institute of Technology Atlanta, GA                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                |
|                                  | <ul style="list-style-type: none"> <li>• Addressed the need for an organization to help Puerto Rican students feel at home and stay connected to our culture</li> <li>• Formed a three-member co-founder board and identified a faculty advisor</li> <li>• Raised funds for the organization and recruited members from campus</li> <li>• Hosted professional, social, and cultural events to meet the needs of the growing Puerto Rican student population at Georgia Tech</li> </ul> |                                                                                                                                                                |
|                                  | <b>Undergraduate Research Mentor</b> August 2021 - May 2022<br>University of Texas at Arlington Arlington, TX                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                |
|                                  | <ul style="list-style-type: none"> <li>• As the lead undergraduate student in Dr. Frank Lewis’ Autonomous Systems Lab, mentored junior undergrads in controls research and guided them on a research project in ground vehicle control</li> <li>• Guided students in Robot Operating System, Python, Linux, and Control Systems</li> </ul>                                                                                                                                             |                                                                                                                                                                |
|                                  | <b>Teaching Assistant</b>   Graduate-Level Intelligent Systems Course January 2022 - May 2022<br>University of Texas at Arlington Arlington, TX                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                |
|                                  | <ul style="list-style-type: none"> <li>• Selected as an undergraduate to TA a graduate-level course</li> <li>• Led exam review sessions and graded homework, exams, and projects</li> </ul>                                                                                                                                                                                                                                                                                            |                                                                                                                                                                |
| <b>HONORS AND AWARDS</b>         | <b>Eta Kappa Nu Public Relations Chair &amp; Circuits I Tutor</b> August 2020 - May 2022<br>University of Texas at Arlington Arlington, TX                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                                                |
|                                  | <ul style="list-style-type: none"> <li>• Executive board member of UT Arlington’s IEEE-HKN chapter</li> <li>• In charge of social media advertising for the honor society’s events and outreach initiatives for the electrical engineering student body</li> <li>• Volunteered to provide free tutoring as a service to electrical engineering students in Circuits I</li> </ul>                                                                                                       |                                                                                                                                                                |
|                                  | <b>Fulbright Specialist</b> August 2026<br>Peer-review-selected to the U.S. Department of State’s Fulbright Specialist Roster for a three-year term                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                |
|                                  | <b>Top-3 Finalist at a Deep Learning Research Symposium</b> November 2023<br>Professor-sponsored award for novel research in deep learning and robotics                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                |
|                                  | <b>Goizueta Fellowship at Georgia Tech</b> August 2022<br>Highly selective fellowship for graduate students of Latin-American descent                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                |
|                                  | <b>Summa Cum Laude Honors at University of Texas at Arlington</b> May 2022<br>Exceptional academic distinction in top GPA tier for graduation honors                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                |
|                                  | <b>Honors College Degree Requirements at University of Texas at Arlington</b> December 2021<br>Completed research projects in disparate fields throughout my undergraduate career with various professors and defended a senior capstone research project before the Honors College committee                                                                                                                                                                                          |                                                                                                                                                                |
|                                  | <b>Electrical Engineering Honors Scholar at University of Texas at Arlington</b> December 2021<br>Inaugural member of the electrical engineering honors cohort                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                |
|                                  | <b>IEEE-HKN International Electrical Engineering Honor Society Membership</b> August 2021<br>Selective undergraduate society recognizing outstanding academic achievement and community service                                                                                                                                                                                                                                                                                        |                                                                                                                                                                |
|                                  | <b>Research Experience for Undergraduates Fellowship Recipient</b> August 2020<br>Fellowship to sponsor promising young undergraduate researchers                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                |

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| <b>Chance Vought Engineering and Science Endowment Scholarship</b>                                 | August 2020  |
| Highly selective award given annually to three recipients across the entire College of Engineering |              |
| <b>Toray Composite Materials America Inc. Sons and Daughters Scholarship</b>                       | May 2019     |
| Academic scholarship awarded to children of employees of Toray America                             |              |
| <b>James E. Casey Scholarship Recipient</b>                                                        | January 2019 |
| Academic scholarship awarded to children of employees of UPS                                       |              |
| <b>James R. Hoffa Memorial Scholarship Recipient</b>                                               | May 2018     |
| Academic scholarship awarded to children of Teamsters Union Members                                |              |
| <b>Grapevine, TX Rotary Club Scholarship Recipient</b>                                             | May 2018     |
| Local rotary club scholarship for academic and civically-minded scholars                           |              |
| <b>National Hispanic Scholar</b>                                                                   | March 2018   |
| Recognition of top national performers of Latin descent on the SAT                                 |              |
| <b>National Merit Scholar</b>                                                                      | March 2018   |
| Recognition of top national performers on the SAT                                                  |              |